Building Science

Lecture 10: Acoustics and Noise Control

Dr. Matthew Francis
School of Property, Construction & Project Management
RMIT University, Melbourne, Australia

RMIT UNIVERSITY

Lecture outline

- Physics of Sound
- Sound measurement
- Noise control in Building design

RMIT UNIVERSITY

Acoustics and Noise Control

Physics of Sound

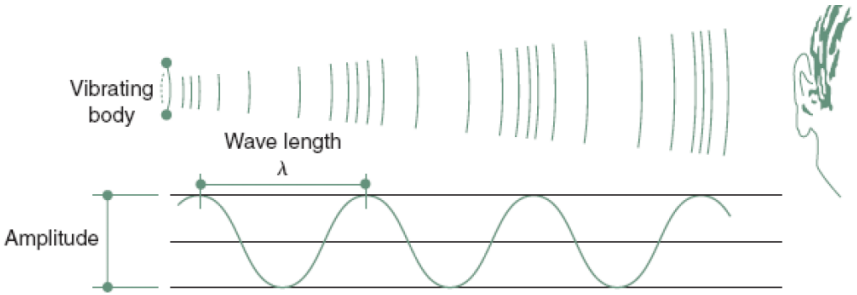
✓Acoustics

✓ The science of sound, of small amplitude mechanical vibrations

✓Sound

✓ A vibration in an elastic medium such as air, water, building materials and the earth

✓ Characterised by wavelength (λ in minutes) or frequency (f in Hertz)



3.1.
Sound waves: longitudinal compression waves and their sinusoidal representation.

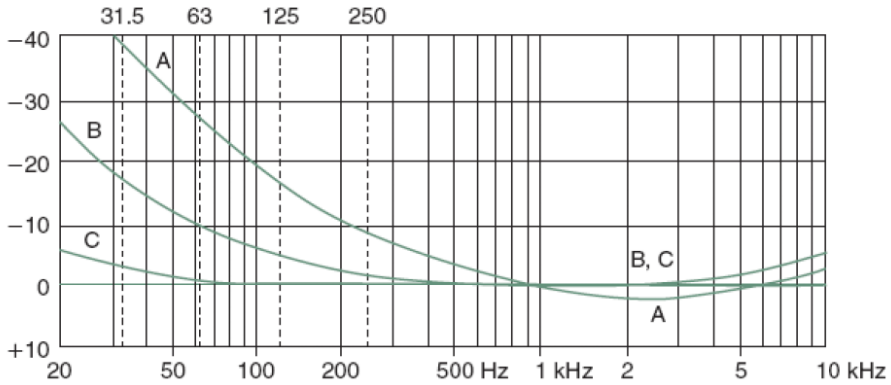
Source: Szokolay, 2008

RMIT UNIVERSITY

Acoustics and Noise Control

dB(A) – a measure like our own ears

- ✓A-weighted Sound level
- ✓One's ability to hear a sound depends greatly on the frequency composition of the sound.
- ✓People hear sounds most readily when the predominant sound energy occurs at frequencies between 1000 and 6000 Hertz (cycles per second)



3.6.
Weighings of sound levels: A, B and C.

RMIT UNIVERSITY

Acoustics and Noise Control

Decibels dB(A) in daily life

Comparison of Intensity, Sound Pressure Level, and Common Sounds

Relative Energy Intensity (units)	*Decibels*[a]	*Loudness*
100,000,000,000,000	140	Jet aircraft and artillery fire
10,000,000,000,000	130	Threshold of pain
1,000,000,000,000	120	
100,000,000,000	110	Near elevated train
10,000,000,000	100	Inside propellor plane
1,000,000,000	90	Full symphony or band
100,000,000	80	Inside auto at high speed
10,000,000	70	
1,000,000	60	Conversation, face-to-face
100,000	50	Inside general office
10,000	40	Inside private office
1,000	30	Inside bedroom
100	20	Inside empty theater
10	10	
1	0	Threshold of hearing

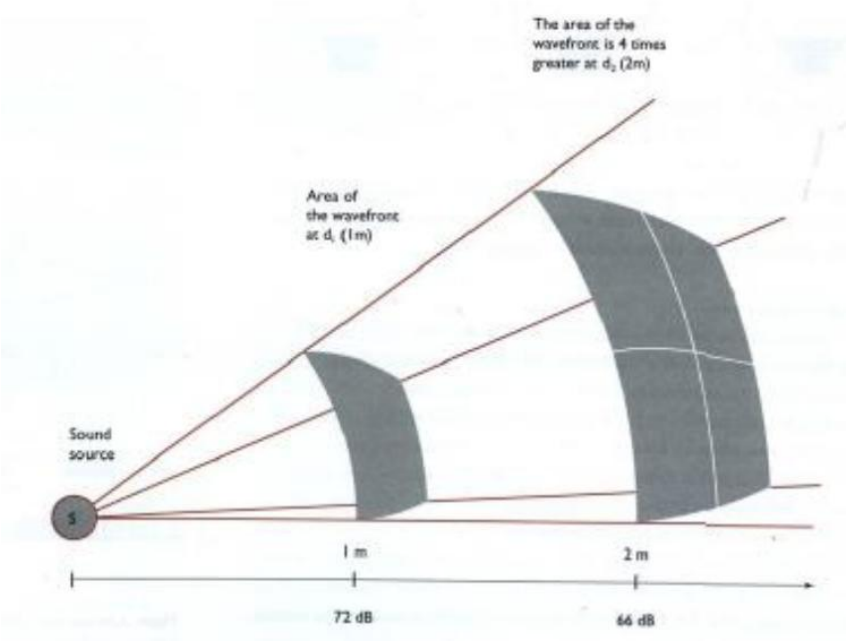
[a]SPL as measured on A-weighted network of standard sound level meter.

RMIT UNIVERSITY

Acoustics and Noise Control

Inverse Square Law

- ✓Decrease in sound pressure as a function of distance
- ✓Each doubling of distance from a reference point translates into a 6 dB loss in the sound pressure level (SPL)



RMIT UNIVERSITY

Perceptions of Sound

RMIT UNIVERSITY

Acoustics and Noise Control

Perception of Sound

- ✓Sound pressure
- ✓The age and health of the hearer
- ✓The frequency of the sound
- ✓The presence of a masking sound

Change of dB Perceptions

Change in dB	Perception of loudness change
3	Barely noticeable
6	Noticeable
10	Twice or half as loud
20	Four times or one-quarter as loud

Source: Lechner, 2012

RMIT UNIVERSITY

Acoustics and Noise Control

Noise Control

- ✓Signals and noise
- ✓Masking effect
- ✓Sound transmission

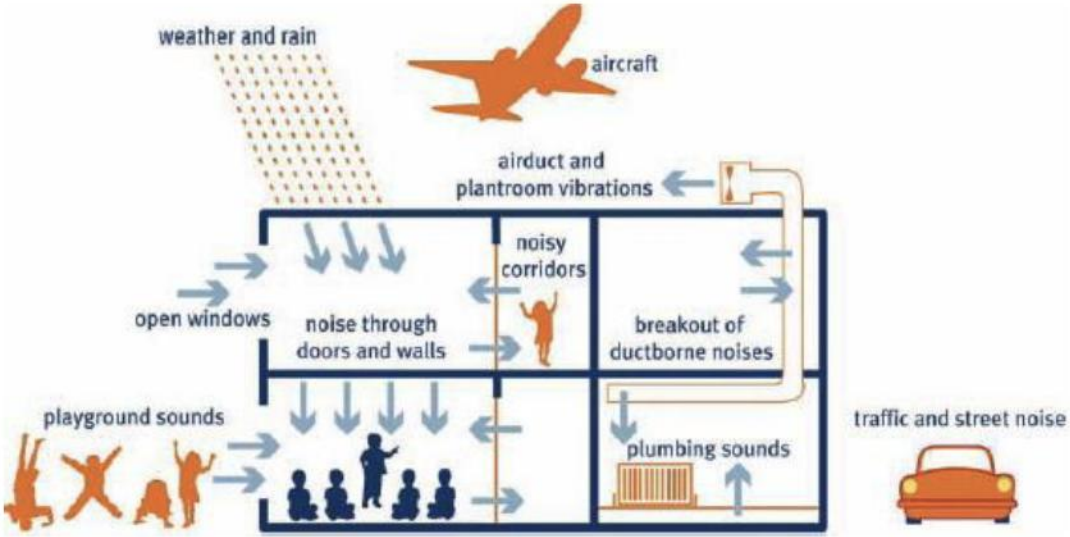


Image: ssc.education.ed.ac.uk

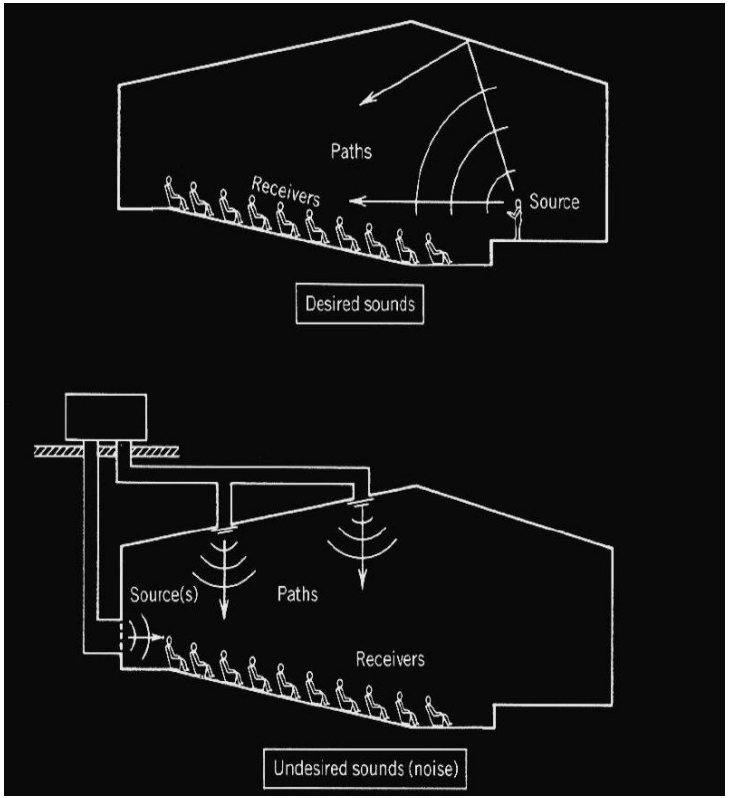
RMIT UNIVERSITY

Acoustics and Noise Control

Signals and Noise

✓Buildings need to maximise
Desired sounds

✓And minimise
Undesired sounds (noise)



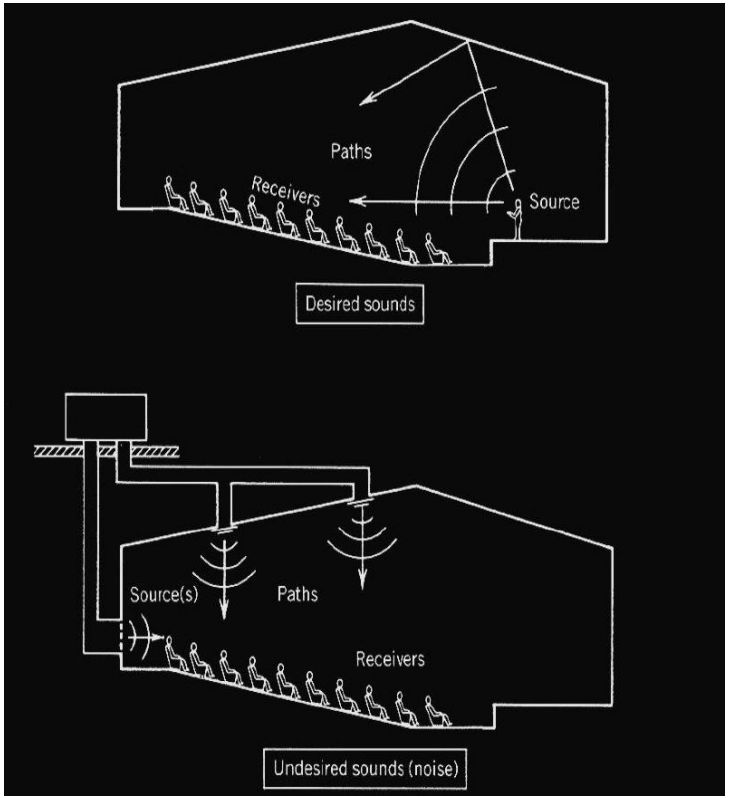
RMIT UNIVERSITY

Acoustics and Noise Control

Signals and Noise

- ✓Any unwanted sound is defined as ***Noise***

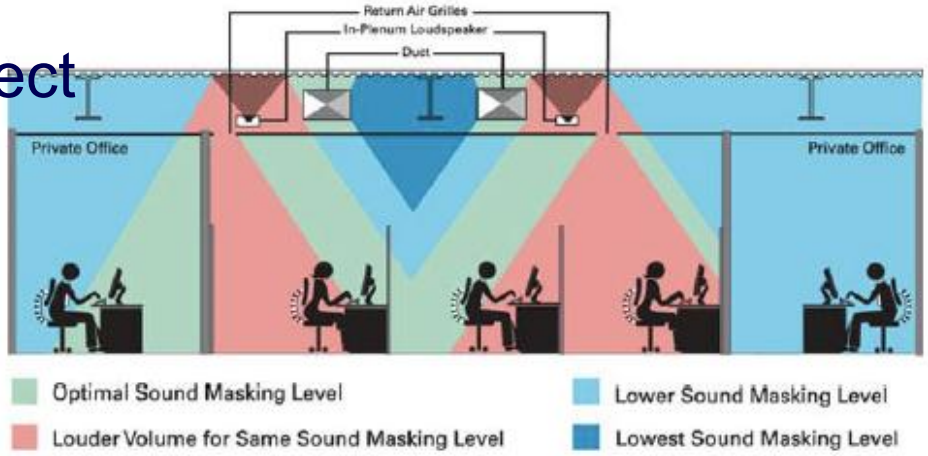
- ✓***Signals*** include sound which communicates wanted information to us. Signal must be presented at a level above the noise if it is to be intelligible (usually + 10 dB)



RMIT UNIVERSITY

Acoustics and Noise Control

Masking effect



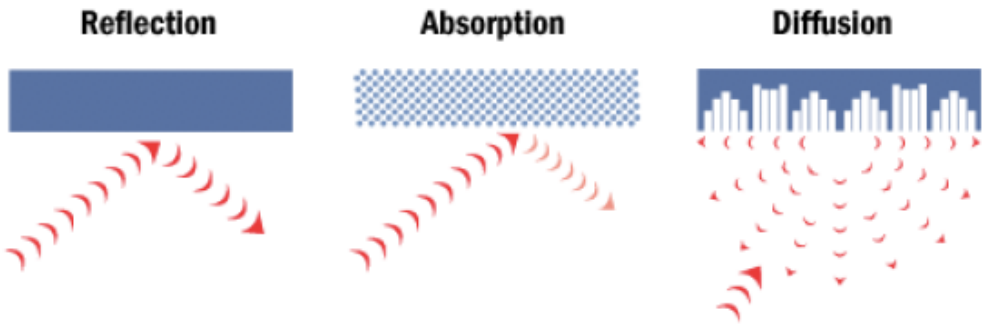
- ✓ **Masking** is when two separate sound sources are perceived simultaneously, the perception of each is made more difficult by the presence of the other

- ✓ Masking is an extremely important and useful technique for noise control, wherein background sound levels are deliberately manipulated to mask other unwanted sounds

RMIT UNIVERSITY

Acoustics and Noise Control

Sound Transmission



- ✓ Reflection
- ✓ Absorption
- ✓ Diffusion

RMIT UNIVERSITY

Acoustics and Noise Control

Noise Reduction Coefficient (NRC)

Category	NRC	Material type
Reflective	0.00	No sound absorption Concrete, smooth brick, still water
	0.05	Rough concrete, normal brick, thick glass, gypsum board, plaster
	0.10	Gypsum boards or wood panels attached to studs or joists
	0.15	Ordinary window glass, thin plywood panel
	0.20	Light carpet
	0.35	Heavy carpet
Absorptive	0.45	Rough soil
	0.55	Heavy carpet on foam rubber pad
	0.60	Heavyweight drapery, grass or gravel
	0.65 to 0.95	Sound-absorbing panels
	0.90	Fresh snow
	1.0	Complete sound absorption

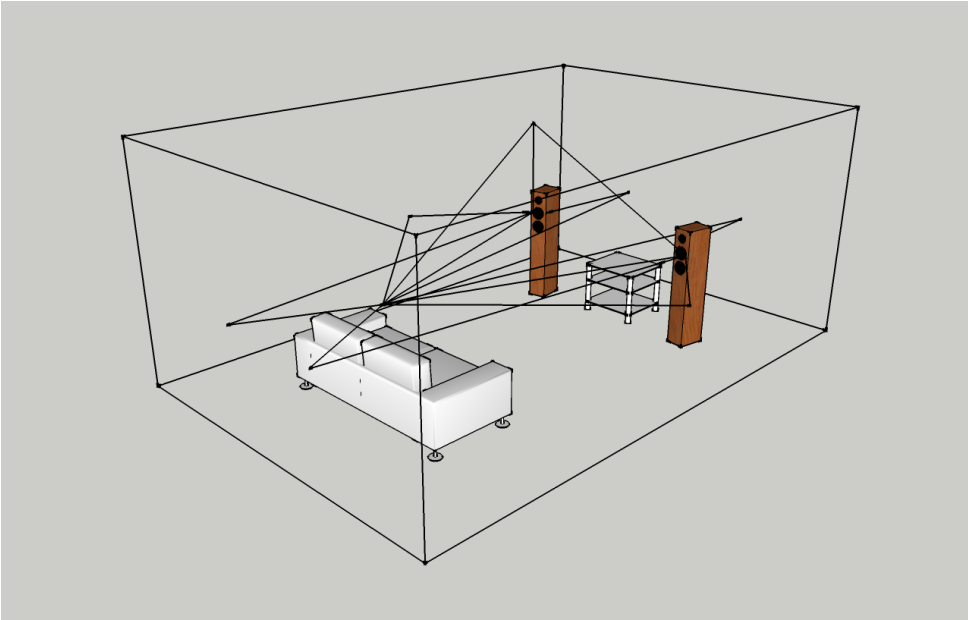
Source: Lechner, 2012

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics

- ✓Requirements for good acoustics
- ✓Room size and shape



RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics (cont'd)

- ✓Diffusion of sound
- ✓Reverberation time
- ✓Reflection
- ✓Warmth and intimacy

RMIT UNIVERSITY

Acoustics and Noise Control

Requirements for good Acoustics

- ✓ *Adequate loudness* – in every part of the room, especially in remote seats
- ✓ The sound energy should be *uniformly distributed* within the room
- ✓ *Optimum reverberation* characteristics for intended uses
- ✓ No *background noise* and vibration

RMIT UNIVERSITY

Acoustics and Noise Control

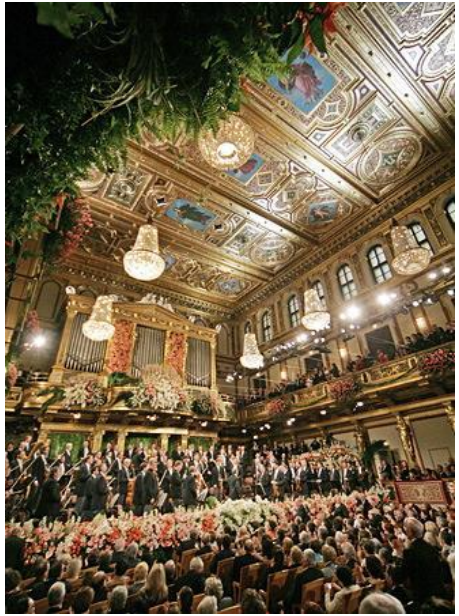
Room Acoustics: Room size

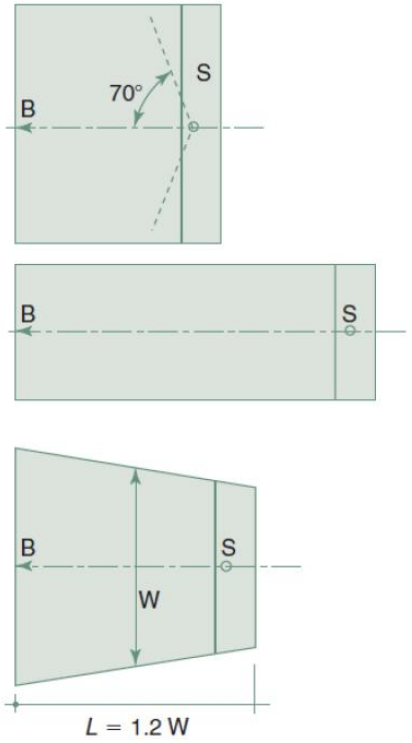
TYPE OF AUDITORIUM	MINIMUM	OPTIMUM	MAXIMUM
Rooms for Speech	$2.3m^3$	$3.1m^3$	$4.3m^3$
Concert Halls	$6.2m^3$	$7.8m^3$	$10.8m^3$
Opera Houses	$4.5m^3$	$5.7m^3$	$7.4m^3$
Catholic Churches	$5.7m^3$	$8.5m^3$	$12.0m^3$
Other Churches	$5.1m^3$	$7.2m^3$	$9.1m^3$
Multipurpose Halls	$5.1m^3$	$7.1m^3$	$8.5m^3$
Cinemas	$2.8m^3$	$3.5m^3$	$5.6m^3$

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics: Room shape





3.47.
Basic issues for shape of auditoria.

Source: Szokolay, 2008

Acoustics and Noise Control

Room Acoustics

- ✓Loudness (strength)
  - ✓Is the volume of the sound loud enough? Is it too loud? Or does it seem louder than it would at the same distance outdoors?
- ✓Clarity
  - ✓Can I hear each of the various instruments clearly? Can I understand what is being sung by a solo vocalist, or what is being said by a speaker?

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics

- ✓Diffusion of Sound
  - ✓To provide a high degree of sound diffusion, room must have surface irregularities, such as ceilings, surface decorations, and deep window reveals.
  - ✓There must also be of various sizes to affect all frequencies

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics

- ✓Reverberation time
  - ✓In an enclosed environment sound can **continue to reflect for a period of time after a source has stopped** emitting sound.
  - ✓This prolongation of sound is called reverberation.
  - ✓Reverberation time (RT60) is defined **as the time required**, in seconds, for the average sound in a room **to decrease by 60 decibels** after a source stops generating sound

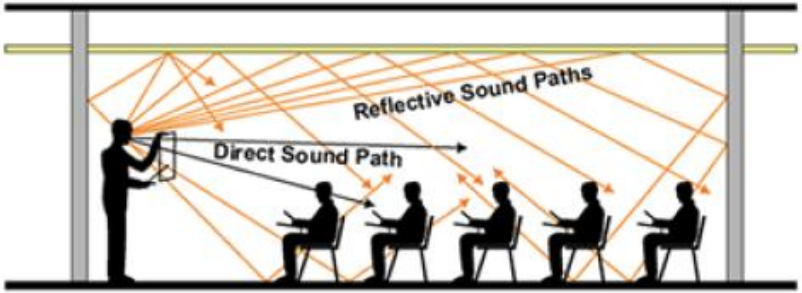


Image source: acousticalsurfaces.com

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics

Reverberation 'Norms'

0.5 Second	+ Recording/Broadcast Studios
1.0 Second	+ Classrooms
	+ Conference/Lecture Rooms
	+ Drama Theaters
1.5 Seconds	+ Chamber Music Halls
	+ Multi-Use Auditoriums
2.0 Seconds	+ Churches (Secular Chorus)
	+ Concert Halls
3.0 Seconds	+ Cathedrals (Pipe Organ / Chant)

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics: Reverberation time

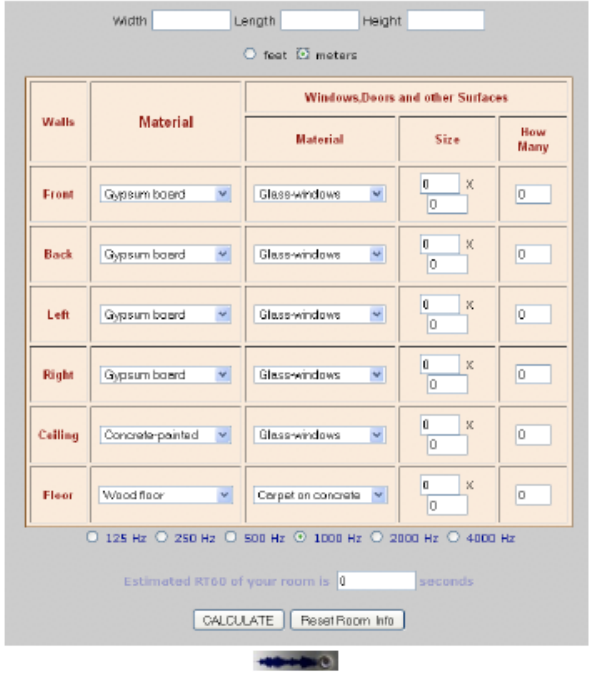
$$RT = 0.16 \times \frac{V}{Abs} \quad \text{(in seconds)}$$

Where V= volume of room (m^3)
Abs = total surface absorption in room (m^2)

RMIT UNIVERSITY

Acoustics and Noise Control

Room Acoustics: Reverberation time



Source: http://www.sae.edu/reference_material/pages/Reverberation%20Time%20Calculator.htm

RMIT UNIVERSITY

Thank you

RMIT UNIVERSITY